



UNIVERSITY OF CAPE TOWN

Financial Analysis and Portfolio Management Final Examination - 12 April 2018 Portfolio Optimization: ECO 4122Z

3 Hours

Maximum Marks: 100

INSTRUCTIONS TO CANDIDATES

- i. Answer ALL questions
- ii. Show all calculations
- iii. Write legibly

1. Suppose on 28 February 2017, you bought 100 shares of stock in African Equity Empowerment Investments Limited for R3.95 per share and you sold them on 28 February 2018 for R7.20 per share. During this period, you received a total cash dividend of R0.108 per share. Compute your HPR and HPY on this African Equity Empowerment Investments Limited stock. [3 marks]
2. Define the following terms:
 - (a) Risk aversion, [2 mark]
 - (b) State-by-state dominance, [2 mark]
 - (c) First-order stochastic dominance. [2 mark]
 [Total 6 marks]
3. Suppose you derive the following information for the companies listed below, based on five years of monthly data:

Company	α_i (Intercept)	σ_i	ρ_{iM}
Alpha	0.25	14.10%	0.72
Beta	0.15	16.60	0.33
Sigma	0.17	8.50	0.55
Lambda	0.05	11.40	0.60
All Share Index (ALSI)	0.00	5.50	1.00

where σ_i is standard deviation for asset i and ρ_{iM} is the correlation between asset i and the market (M).

- (a) Compute the beta coefficient for each stock. Comment on whether each stock is aggressive or defensive. [4 marks]
 - (b) Assuming a risk-free rate of 8 percent and an expected return for the market portfolio of 15 percent, compute the expected (required) return for all the stocks and plot them on the Security Market Line (SML) and state whether the above securities are over- or under-valued. [5 marks]
 - (c) Suppose two weeks after you have plotted the SML above, the South African Reserve Bank (SARB) revises its inflation forecasts downward by 5 basis points. Explain the likely impact of this on the SML (you may use a graph to illustrate your answer). [4 marks]
- [Total 13 marks]
4. Suppose the market price of GNMILLER, a company listed in the JSE is R400 per share, the expected return is 13%, and the riskless rate of interest is 7%, and the market risk premium, $[E(R_m) - R_f]$ is 8%. What will be GNMILLER's current price if its expected

future payoff remains the same but the covariance of its rate of return with the market portfolio increases by one-and-a-half times? (i.e. from $\mathbf{V}$ to $\mathbf{V} \times 1.5$) [4 marks]

5. Suppose you have been assigned the task of estimating the expected returns for three different stocks: ABC, DEF, and GHI. Your preliminary analysis has established the historical risk premiums associated with three risk factors that could potentially be included in your calculations: the excess return on a proxy for the market portfolio (Mkt), and two variables capturing general macroeconomic exposures ($Macro1$ and $Macro2$). These values are: $\lambda_{Mkt} = 6.52\%$, $\lambda_{Macro1} = 0.35\%$, and $\lambda_{Macro2} = 0.62\%$. You have also estimated the following factor betas (i.e., loadings) for all three stocks with respect to each of these potential risk factors:

Stock	Factor Loading		
	Mkt	Macro1	Macro2
ABC	1.25	-0.52	0.00
DEF	0.91	0.64	0.32
GHI	1.03	-0.09	0.00

- (a) Calculate expected returns for the three stocks using just the Mkt risk factor. Assume a risk-free rate of 4.5%. [3 marks]
- (b) Calculate the expected returns for the three stocks using all three risk factors and the same 4.5% risk-free rate. [3 marks]
- (c) Discuss the differences between the expected return estimates from the single-factor model and those from the multifactor model. In the context of South African financial markets, which estimates are most likely to be more useful? [5 marks]
- (d) What sort of exposure might $Macro2$ represent? Given the estimated factor betas, is it really reasonable to consider it a common (i.e., systematic) risk factor? [3 marks]

[Total:14 marks]

6. With regard to market efficiency, what is meant by the term “anomaly”? Using any two examples market anomalies in South Africa, explain how a portfolio manager can use her understanding of these anomalies to ‘beat’ the market. [5 marks]

7. You are given the following information regarding prices for a sample of stocks.

Day	Price (Rand)			Shares (millions)		
	A	B	C	A	B	C
27 March	220	330	620	1000	700	500
28 March	200	320	650	1000	700	500
29 March	240	560	620	1000	350*	500
30 March	230	570	350	1000	350	1000*
31 March	220	550	360	1000	350	1000

*There was a stock split at the close of 28 March and at the close of 29 March

- (a) Construct a price-weighted index for these three stocks, and compute the percentage change in the index for the period from 27 March to 31 March. Set your initial divisor at 10. [9 marks]
- (b) Suppose after calculating the index above, one of your colleagues argues that you should have used a market-value weighted index. Explaining the difference between the methods of index construction, outline the advantages and disadvantages of each method. [6 marks]
- (c) The Johannesburg Stock Exchange (JSE) uses a free-float factor weighting approach to index construction. Outline the main advantages of this approach. [4 marks]

[Total: 19 marks]

8. You are evaluating the performance of two portfolio managers, and you have gathered annual return data for the past decade as follows:

Year	Manager A	Manager B
	Return (%)	Return (%)
1	-6.5	-9.8
2	3.6	-8.7
3	4.8	3.3
4	8.4	8.6
5	13.8	12.7
6	-6.2	8.6
7	-5.4	9.2
8	9.5	11.2
9	11.3	12.8
10	13.6	13.1
11	18.7	17.6
12	22.4	18.2

- (a) For each manager, calculate (i) the average annual return, (ii) the standard deviation of returns, and (iii) the semi-deviation of returns. [6 marks]

- (b) Assuming that the average annual risk-free rate during the sample period was 3.8 percent, calculate the Sharpe ratio for each portfolio. Based on these computations, which manager appears to have performed the best? [3 marks]
- (c) Calculate the Sortino ratio for each portfolio, using the average risk-free rate as the minimum acceptable return threshold. Based on these computations, which manager appears to have performed the best? [3 marks]
- (d) When would you expect the Sharpe and Sortino measures to provide (i) the same performance ranking, or (ii) different performance rankings? Explain. [3 marks]
- (e) Suppose after your computations above, one of your clients asks you to distinguish among the following alternative performance measures: The Treynor measure, Jensen's alpha, and M^2 (also known as the risk-adjusted performance measure). State whether each of these alternative measures assumes that the relevant risk is systematic, unsystematic, or total. Outline the main advantage of each measure. [3, 3 marks]

[Total: 21 marks]

9. Consider the following historical performance data for two different portfolios, the JSE All Share Index, and the 90-day T-bill.

Investment	Average Return	Standard Deviation	Beta	R^2
Fund 1	26.40%	20.67%	1.351	0.751
Fund 2	13.22	14.20	0.905	0.713
JSE All Share	15.71	13.25		
90-day T-bill	6.20	0.50		

- (a) Calculate the Fama (1972) overall performance measure for both funds. [2 marks]
- (b) What is the return to risk for each funds? [2 marks]
- (c) For each funds, compute the measures of (i) selectivity, (ii) diversification, and (iii) net selectivity. [6 marks]
- (d) Explain the meaning of the net selectivity and diversification measure and how they help you evaluate investor performance. Based on these, which fund had the best performance? [5 marks]

[Total: 15 marks]